AdaptiveOcc: Adaptive Octree-Based Network for Multi-Camera 3D Semantic Occupancy Prediction in Autonomous Driving

Tianyu Yang, Yeqiang Qian, *Member, IEEE*, Weihao Yan, Chunxiang Wang, *Member, IEEE*, and Ming Yang, *Member, IEEE*

***Abstract*— Multi-camera 3D semantic occupancy prediction is a critical task for autonomous driving, playing a vital role in understanding the environment. Current methods mainly rely on uniform voxel representation to encode space, which greatly limits their resolution scalability. It causes most existing methods to struggle with scaling to finer granularities, as the cubic growth nature of uniform voxel leads to a significant increase in the demand for computational and storage resources when scaling. To address this, we propose a multi-level hierarchical model AdaptiveOcc. Using the octree structure, our model can adaptively represent different parts of space with varying voxel granularity. It can selectively extend resolution only for a small subset of voxels, thus mitigating the substantial computational and storage burden brought by scaling. To endow our model with adaptability, we propose a distance-adaptive octree construction rule for generating supervised labels. Considering that the voxel granularity requirements vary for different distance ranges in environmental perception, such a construction rule results in a higher likelihood of coarser granularity for distant regions and finer granularity for nearby regions. This ensures a more efficient and rational allocation of computational resources, further reducing the inference latency. Extensive experiments on nuScenes, SemanticKITTI and Waymo dataset validate that our method can scale to finer granularities with faster speed, and less training memory compared with other state-of-the-art methods. Our code is available at https://github.com/yty-sky/AdaptiveOcc.**

***Index Terms*— Multi-camera 3D semantic occupancy prediction, adaptive octree-based network, resolution scalability, distance-adaptive octree construction rule.**

I. Introduction

With the evolving demands for more refined and holistic visual perception capabilities in autonomous driving, 3D semantic occupancy prediction has emerged. It requires predicting both the occupancy status and semantic labels of each voxel in 3D space from multi-view images. Compared to traditional 3D object detection [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], 3D semantic occupancy has the capability to perceive both the foreground and background and provides more fine-grained geometric perception of object shapes, rather than the coarse-grained 3D bounding box. Furthermore, 3D semantic occupancy networks have the potential to infer occluded spaces by leveraging surrounding semantic cues, which is unavailable in depth estimation [13], [14], [15], [16], [17], [18], [19]. Therefore, it is considered a valuable perceptual representation.

There are some pioneering methods exploring the realization of this task. MonoScene [20] built the first monocular method for 3D semantic occupancy prediction. It constructs the uniform voxel feature by line-of-sight projection and employs the 3D UNet to handle them. Focused on multi-camera implementation, SurroundOcc [21] proposes a scheme for generating dense ground truths and adopts a multi-scale architecture, which utilizes multi-scale query to aggregate 2D features into 3D voxel features. FB-OCC [22] proposes a joint projection approach to acquire voxel features and leverage a stronger large-scale backbone and more efficient pre-training techniques, achieving significant breakthroughs in accuracy. Although these methods have successfully achieved the task requirements of 3D semantic occupancy prediction, the granularity of predicted voxels in these methods is relatively coarse, generally around 0.5 meters. This level of granularity is too coarse for certain driving scenarios, especially in scenarios requiring fine-grained perception at short-distance ranges, such as overtaking, parking, and driving on congested roads. Moreover, the resolution scalability of these existing methods [20], [21], [22] is limited due to their reliance on uniform voxel representation when encoding space. This makes it challenging for them to scale to finer granularities, as the cubic growth nature of uniform voxel leads to a significant increase in the demand for computational and storage resources when scaling. Specifically, when scaling from 0.5m granularity to 0.25m, the state-of-the-art method, SurroundOcc [21], experiences a significant increase of 115.2% in inference latency and 123.8% in training memory. Such limitation greatly constrains the practical application of these models. To this point, our aim is to develop an efficient method with strong resolution scalability.

To address this, we propose a multi-level hierarchical model AdaptiveOcc. AdaptiveOcc utilizes the octree structure for feature propagation and output, adaptively representing different parts of the space with varying voxel granularity. It can

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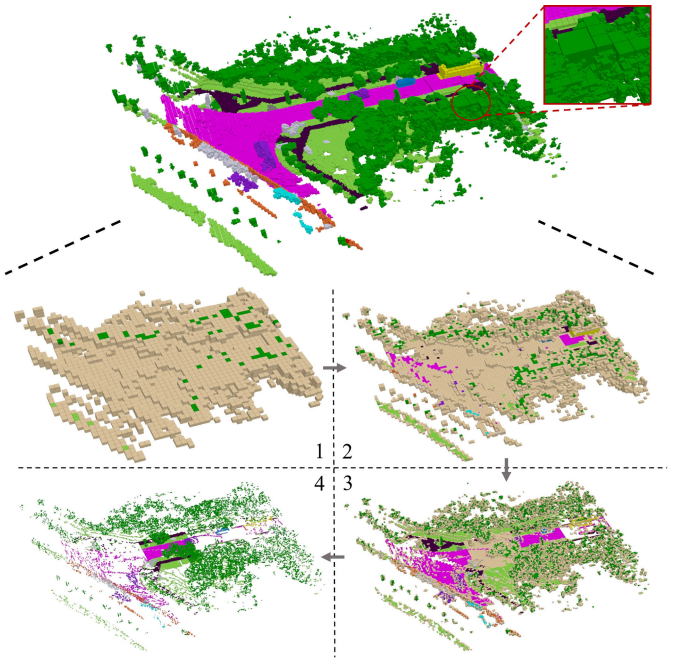


Fig. 1. **Octree-based multi-level hierarchical structure.** The scene is represented by multi-level voxels, transitioning from coarse to fine granularities. Each layer consists of determined voxels and indetermined voxels, with indetermined voxels marked by beige. Except for the first layer, the voxels in each subsequent layer are obtained through octree subdivision of the indetermined voxels from the previous layer. At the top of the image is the merged representation of the multi-level voxels for the entire scene, eliminating indetermined voxels.

selectively extend the resolution only on a small portion of voxels, thereby avoiding the significant computational and storage burden associated with uniform voxel representation when scaling. From a network architecture perspective, our model adopts a multi-layered output pattern, where the granularity of the output voxels becomes progressively finer from shallow to deep layers. Large homogenous non-empty voxels and empty voxels can be easily identifiable, requiring only the superficial layers of the network for differentiation without delving into its deeper layers. These large voxels are processed and outputted in the early layers, sparing subsequent computational and storage resources. Only intricate structures in space that are challenging to differentiate are retained for the deeper layers of the network and are represented using finer granularity voxels.

To endow our model with adaptability, it is crucial to construct corresponding supervised labels to guide the output of the network. We represent the scene using a multi-level hierarchical structure implemented by the octree [23], as illustrated in Figure 1. Within the multi-level hierarchical structure, non-empty voxels at each layer are categorized into two states: determined voxels and indetermined voxels. Voxels in different states will guide the network to take different actions. Determined voxels at a certain layer need to be directly output from the network, while indetermined voxels, after octree subdivision, are passed on to the next layer for further processing.

To make full use of the hierarchical structure of our model and further accelerate the inference speed, we propose a distance-adaptive octree construction rule to distinguish between determined and indetermined voxels. We distinguish them based on intra-voxel semantic statistics and custom rule restrictions. Considering that the voxel granularity requirements vary for different distance ranges in environmental perception, we relatively relax the restrictions on merging voxels into determined voxels for both medium and long-distance ranges, which is the core of distance adaptability. Such a distance-adaptive strategy results in a higher likelihood of coarser voxel granularity for distant regions and finer voxel granularity for nearby regions. This distribution pattern guarantees a more efficient and rational allocation of computational resources, further reducing the inference latency and storage demands.

Finally, we validated our method on the nuScenes [24], SemanticKITTI [25] and Waymo [26] dataset, demonstrating superior performance in terms of resolution scalability. Our method can scale to finer granularities with faster speed, and less training memory compared with other state-of-the-art methods. The main contributions of this paper are as follows:

- We propose a novel octree-based multi-level hierarchical model designed for multi-camera 3D semantic occupancy prediction, demonstrating strong resolution scalability. It can adaptively represent different parts of the space with varying voxel granularity and selectively extend the resolution only on a small portion of voxels when scaling.
- We propose a distance-adaptive octree construction rule for generating supervised labels to endow our model with adaptability. This enables a more efficient and rational allocation of computational resources, further reducing inference latency and storage requirements.
- Extensive experiments on nuScenes, SemanticKITTI and Waymo dataset validate that our method can scale to finer granularities with faster speed, and less training memory compared with other state-of-the-art methods.

II. Related Work

A. Camera-Based BEV Perception

Considering the dimension gap between the 2D image input and the 3D prediction, camera-based BEV perception [2], [3], [27], [28], [29], [30], [31], [32], [33] has become a standard paradigm, particularly for surround-view camera scenarios. It first constructs the BEV feature representations through projection, and then performs various downstream tasks on the BEV space. Based on the feature projection approach, it can be mainly categorized into three methods. The forward projection method [4], [27], [34], [35], exemplified by LSS [4], predicts the pixel-wise depth distribution by a dedicated depth estimation network, eventually obtaining BEV features through grid sampling. Owing to the utilization of explicit depth estimation, such methods possess strong geometric perception capabilities and high training efficiency. The backward projection method [2], [36], exemplified by BEVFormer [2], treats each cell in the BEV space as a query and utilizes the deformable attention [37] to update them with corresponding image features, which exhibits enhanced flexibility in feature acquisition. The joint projection method [38]

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uses forward projection to generate the initial BEV features. Then the BEV features are treated as queries within the BEV space and attend the image encoder features to acquire dense geometry information. The method combines the advantages of both forward and backward projection techniques, resulting in greater accuracy. Our work adopts the backward projection method to extract features from 2D images. Forward projection is the process of pushing 2D image features into 3D space, typically bound with a uniform grid feature representation. Because its fixed voxel granularity simplifies the determination of the number of bins required for depth distribution and the simple, continuous structure facilitates grid sampling. Our work adopts an octree structure to represent features, which is irregular and allows for variable voxel granularity, making it challenging to efficiently map image features onto this structure through forward projection. However, backward projection is the process of pulling features from 2D images onto 3D space. By utilizing intrinsic and extrinsic parameters, we can easily determine the corresponding regions in the image for each grid in the irregular octree structure, facilitating efficient feature retrieval. Its highly flexible feature aggregation capability is well-suited for the adaptive octree structure in this study.

B. Vision-Based 3D Semantic Occupancy Prediction

Vision-based 3D semantic occupancy prediction, which requires predicting the occupancy status and semantic labels of each voxel in 3D space through the camera, is an important task in autonomous driving. It can be divided into monocular [20], [39], [40] and multi-camera methods [21], [22], [41], [42], [43]. For the monocular methods, MonoScene [20] builds the first monocular method to infer the dense 3D voxelized semantic scene, which employs the 3D UNet to handle voxel features produced by line-of-sight projection. OccFormer [39] employs a backward projection approach to generate voxel features and then uses a dual-path transformer network, decomposing the heavy 3D processing into local and global transformer pathways along the horizontal plane. For the multi-camera methods, TPVFormer [41] introduces a tri-perspective view representation to describe the scene, straightforward but limited in its feature expressiveness. SurroundOcc [21] proposes a scheme for generating dense ground truths and adapts a multi-scale architecture, which utilizes multi-scale query to aggregate 2D features into the 3D features. This work is comprehensive and solid, making significant contributions both in dense ground truths generation and network model, which serves as the baseline for our work. FB-OCC [22], MiLO [42], and UniOcc [43], the top three solutions presented at the CVPR workshop, all adopt a variety of strategies, such as stronger backbone, joint training of semantics and depth, model ensemble, long-term temporal, test-time augmentation, and so on. While these techniques have achieved remarkable breakthroughs in accuracy, their demands for computational and storage resources are substantial, making them challenging to deploy in real-world autonomous driving scenarios.

C. Octree-Based Convolutional Neural Networks

The octree technique [23], [44], [45] partitions a three-dimensional space recursively by subdividing it into eight octants. It serves as a central technique for many computer graphics applications, like rendering, shape reconstruction, and collision detection. Due to its spatial efficiency and friendliness to GPU implementation, the octree representation is inherently well-suited for the 3D semantic occupancy prediction task. O-CNN [23] was the first to propose the octree-based neural network for 3D shape analysis. This work implements many standard operators such as convolution, normalization, pooling, and deconvolution on the octree, which brought significant performance improvements in terms of memory and computational cost. The subsequent research, Adaptive O-CNN [45], can adaptively estimate the best plane normal and displacement across different layers for different shape analysis and generation tasks, which inspires our work.

III. Approach

A. Problem Formulation

The multi-camera 3D semantic occupancy prediction requires predicting the 3D semantic occupancy of surrounding scenes with multi-camera images $I = \{I^1, I^2, \cdots I^N\}$. The surrounding scenes are discretized into a voxel space of dimensions (H, W, Z) based on the voxel granularity. It can be formalized as follows:

$$V = N(I^1, I^2, \cdots I^N) \tag{1}$$

where N is an neural network and $V \in \mathbb{R}^{H\times W\times Z}$ is the 3D occupancy. The value of V is between 0 and 1, representing the occupied probability of the voxels. By elevating V to a tensor of dimensions (H, W, Z, L), we can obtain the 3D semantic occupancy, where L is the class number and class 0 means non-occupied grids.

B. Overview

Figure 2 shows the pipeline of our method. Given a set of surrounding multi-camera images, we initially employ a image backbone network (e.g. ResNet-101 [46]) to extract N cameras' and M levels' multi-scale features $X = \{\{X_i^j\}_{i=1}^N\}_{j=1}^M$, followed by passing these multi-scale features into the adaptive octree-based module. For each level, we use a transformer to update 3D volume queries by aggregating 2D multi-scale image features through spatial cross attention. Then the adaptive octree-based module evaluates the state of each voxel at each layer and classifies them into empty, determined, and indetermined voxels. The empty and determined voxels are directly outputted and do not participate in subsequent network propagation. While indetermined voxels are split into eight smaller voxels for processing by the subsequent layers. Ultimately, multi-layer results are generated, which can be aggregated to derive the complete 3D occupancy and semantics of the entire surrounding space.

To train our model and endow it with adaptability, we propose a distance-adaptive octree construction rule for generating supervised labels. The rule assigns distinct construction strategies based on varying distance ranges from the sensor, which

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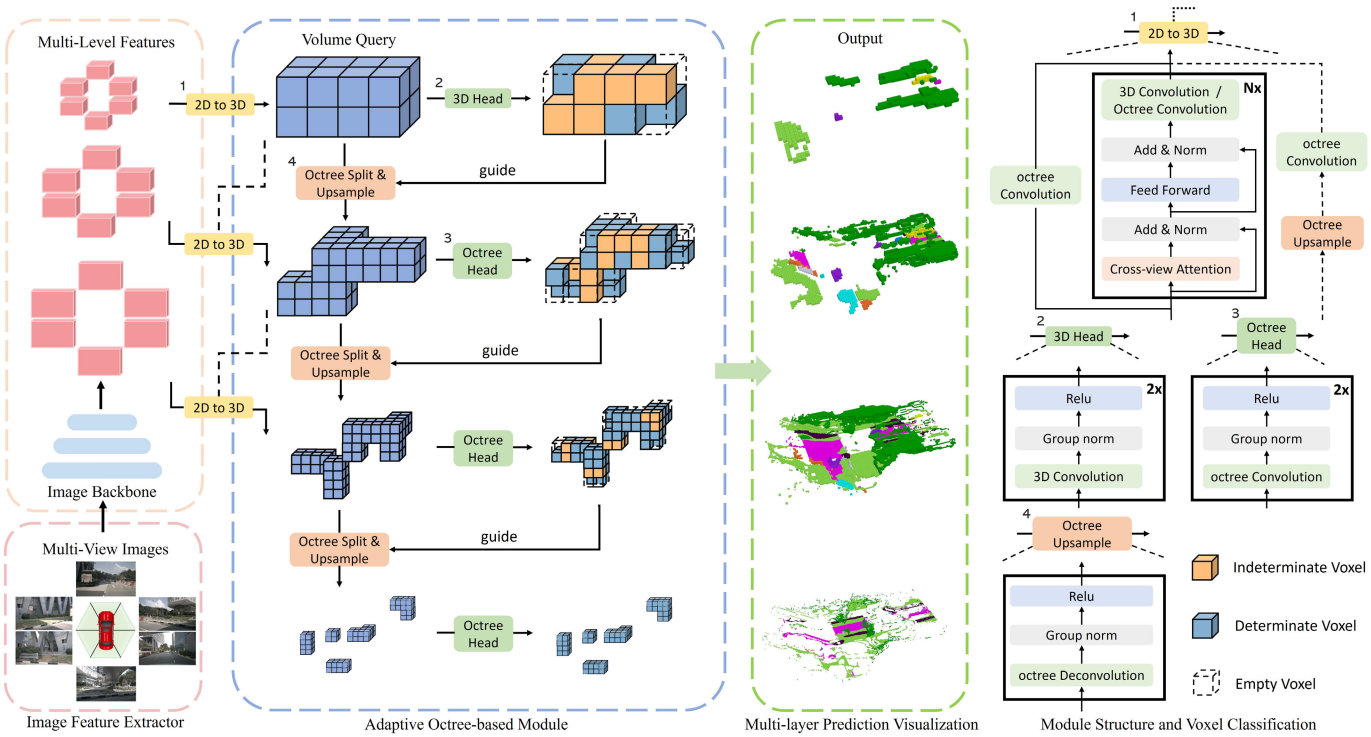


Fig. 2. **The pipeline of AdaptiveOcc**. First, we use a backbone network to extract multi-scale image features, and then employ the 2D to 3D transformer module to convert these 2D image features into 3D voxel features. Subsequently, the adaptive octree-based module utilizes the octree head or 3D head to adaptively classify voxels into empty, determined, and indetermined voxels at each layer. Empty and determined voxels are directly outputted, while indetermined voxels are split into eight smaller voxels for processing by the subsequent layers until reaching the last layer. The octree upsample module is responsible for upsampling the features of an indeterminate voxel into eight voxel features. Ultimately, multi-layer prediction results are generated and can be aggregated to derive the complete 3D occupancy and semantics of the entire surrounding space.

results in a higher likelihood of coarser voxel granularity for distant regions and finer voxel granularity for nearby regions. The output of the adaptive octree-based module is supervised by the constructed supervised labels layer by layer. Furthermore, we introduce an adaptive loss to mitigate the issue of class imbalance, which assigns greater weight to minority and challenging semantic classes.

C. Adaptive Octree-Based Module

The adaptive octree-based module serves as the core of the entire framework. It can adaptively output large homogeneous voxel blocks at shallower layers while ensuring that intricate voxel structures are relayed at deeper layers. There are some important definitions to note before delving into this module. Within the output of each layer of this module, voxels can be categorized into three distinct states:

- Indetermined voxel: These voxels are non-empty but the granularity is too coarse to provide a detailed representation of the spatial region they occupy, thereby needing further subdivision. These voxels are regarded as having ambiguous semantics.
- Determined voxel: These voxels are non-empty and the granularity is suitable for accurately representing the spatial region they occupy. These voxels contain specific semantics, such as car, pedestrian, bicycle, etc.
- Empty voxel: These voxels are empty.

1) Query Points and Aggregate Information: From the perspective of network flow, this module maintains an initial volume query denoted as $Q_0 \in \mathbb{R}^{H\times W\times Z\times C_0}$. We use a transformer to update volume query Q_0 by aggregate information from multi-scale image features, which is shown in Figure 2 as "2D to 3D". The cross-view attention layer constitutes the central component of the "2D to 3D" transformation, which projects 3D reference points to 2D views and uses deformable attention [2], [37] to aggregate information. Specifically, utilizing the provided intrinsic and extrinsic parameters, we project each designated query's 3D point onto its corresponding 2D view. Then we sample 2D features around the projected 2D positions and perform a weighted summation of them, using only the views hit by the 3D reference point. The output $F \in \mathbb{R}^{H\times W\times Z\times C_0}$ of the cross-view attention layer can be formulated as follows:

$$\begin{aligned} &DeformAttn(q, p, x) \\ &= \sum_{i=1}^{N_{head}} W_i \sum_{j=1}^{N_{key}} A_{ij} \cdot W_i' x(p + \Delta p_{ij}) \\ &F^p = \frac{1}{|V_{hit}|} \sum_{i\in V_{hit}} DeformAttn(Q^p, P(q^p, i), X_i) \end{aligned} \tag{2}$$

where F^p and Q^p indicate the pth element of the output features and volume queries. q^p is the corresponding 3D positions of queries and P is the 3D to 2D project function. V_{hit} represents the hit views of 3D query points. After passing through several layers of cross-view attention and

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other conventional transformer components, we obtain the aggregated feature $\overline{Q}_0 \in \mathbb{R}^{H\times W\times Z\times C_0}$.

2) Evaulate and Split Voxels: Subsequently, the aggregated feature $\overline{Q}_0$ is passed to a 3D head, which consists of 3D convolution, group normalization, and relu activation, to evaluate the state of each voxel. If a voxel is classified as a determinate voxel or an empty voxel, its occupancy and semantics result will be directly output at the current layer. Otherwise, it will be passed to the next layer as an indetermined voxel for further processing. Assuming that there are N voxels classified as indeterminate voxels in this layer, we perform an octree split on these N voxels and upsample at the corresponding locations in the aggregated feature $\overline{Q}_0$, to obtain a new octree volume query denoted as $Q_1 \in \mathbb{R}^{8N\times C_1}$. The role of the octree upsample module is to upsample the features of an indeterminate voxel into eight voxel features, thus connecting adjacent layers of the network.

3) Double Skip Connection: After obtaining the new octree volume query Q_1, we aggregate features on a higher resolution feature map to produce the output $\overline{Q}_1$:

$$\overline{Q}_i = Transformer(Q_i, X_i) + Conv(Q_i) + Conv(Upsample(Q_{i-1})) \quad (3)$$

where $Transformer(\cdot)$ indicates several layers of cross-view attention and other conventional transformer components. $Conv(\cdot)$ indicates octree convolution and $Upsample(\cdot)$ indicates octree upsample. Additionally, it's worth noting that we not only gather information from the 2D multi-camera feature maps but also fuse the query features from the current and previous layers in a double skip-connection manner. This double skip-connection manner enriches the features expressiveness, endowing them with both coarse-grained information from the previous layers and fine-grained information from the current layer, resulting in improved prediction results.

4) Iterate and Output: Following the same pattern as described above, we feed $\overline{Q}_1$ into the octree head, evaluate and split voxels, and then pass them to the next layer. It's worth noting that the feature arrangement at this stage is no longer a uniform grid structure but an octree necessitating the substitution of the 3D convolution in the head with octree convolution. This process iterates until the final layer, with outputs from coarse-grained to fine-grained. To conserve computational resources, the feature in the final layer is obtained directly through upsampling from the previous layer and the output from the final layer consists of only determined voxels and empty voxels. Notably, there exists a correspondence when converting multi-scale image features into multi-level voxel features. Deeper image features have larger receptive fields and contain richer semantic information, whereas shallower image features have smaller receptive fields with more detailed information. To handle and differentiates different types of voxels at various levels of granularity, we correspond deeper image features with coarser-grained voxel features, and shallower image features with finer-grained voxel features. That's because coarser-grained voxel features cover larger spatial areas and potentially encompass multiple semantic categories, thus requiring larger receptive fields and richer semantic information. Fine-grained voxel features need to pay more attention to local geometric information to capture finer spatial details. Based on this correspondence, the model outputs easily recognizable large empty voxels and homogeneous non-empty voxels at shallow layers. Voxels that are difficult to identify or have fine structural details are considered indeterminate voxels and passed to the next layer. And the model outputs fine-grained edge structures at deep layers.

D. Distance-Adaptive Octree Construction Rule

To endow our model with adaptability and make full use of the hierarchical structure, it is essential to establish corresponding supervised labels for training. We need to construct multi-layered supervised labels and define rules for each layer to distinguish between indetermined and determined voxels.

Initially, we adopt the scheme proposed by Wei et al. [21] to generate dense occupancy ground truth with a resolution of 0.25 meters, denoted as $G_{input} = \{x_i, y_i, z_i, s_i\}_{i=1}^{N} \in \mathbb{R}^{4\times N}$. G_{input} represents the set of all N non-empty voxels, x_i, y_i and z_i are the three-dimensional spatial coordinates of ith non-empty voxels, and s_i denotes its semantics. Assuming the number of layers in our adaptive octree-based module is l, denoted as $level = l$. To align with the adaptive octree-based module, we need to construct l-layer supervised labels $G_{output} = \{\{x_i^j, y_i^j, z_i^j, s_i^j\}_{i=1}^{N_i}\}_{j=1}^{l}$, where $s_i^j (j < l)$ introduces an additional ambiguous semantics to represent indetermined voxels. And the generation of indetermined voxels in the entire multi-layered labels is controlled by the $Rule = \{\delta_1, \delta_2, \delta_3\}$. The overall algorithm is illustrated in Algorithm 1.

We build the labels from the top layer l to the bottom layer. We first convert the G_{input} to G_{output}^{l} by a direct replication, obtaining the initial labels for lth layer of G_{output}. Then based on the spatial coordinate relationships, we treat adjacent voxels within an octree node as a cell. We conduct semantics voting among the voxels inside a cell to determine the most prevalent semantic label s^* and its corresponding vote count cnt. The $Rule = \{\delta_1, \delta_2, \delta_3\}$ varies with the distance range and satisfies $\delta_1 \geq \delta_2 \geq \delta_3$. We determine the distance range in which each cell resides based on its spatial coordinates and obtain its corresponding rule parameter δ^*. Specifically, δ^* is set to δ_1 for cells in the short-distance range ($0 \sim 10m$), to δ_2 for those in the mid-distance range ($10 \sim 30m$), and to δ_3 for those in the long-distance range ($30 \sim 50m$). For each cell, If the vote count cnt is greater than the rule parameter δ^*, and the most prevalent semantics s^* is not the ambiguous semantics, it can be inferred that these small voxels within the cell can be merged into one larger voxel. In this case, the semantics of the cell is set to s^*. Otherwise, the cell will be assigned ambiguous semantics. Figure 3 illustrates some cases to facilitate a better understanding of this process. Once the semantics for all cells are determined, the construction of G_{output}^{l-1} is completed.

Following the same procedure, we iteratively construct the supervised labels until we reach the first layer. It's important to note that as we complete the construction of one layer, our rule parameters need to become stricter. This is due to the fact that as the layer becomes lower, the granularity of voxels becomes coarser, and employing the same rule parameters might lead to a larger discrepancy with the actual occupancy

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Algorithm 1 Distance-Adaptive Supervised Label Generation

Input:

$G_{input} = \{x_i, y_i, z_i, s_i\}_{i=1}^{N} \in \mathbb{R}^{4\times N}$

$Level = l$

$Rule = \{\delta_1, \delta_2, \delta_3\},\ \delta_1 \geq \delta_2 \geq \delta_3$

Output:

$G_{output} = \{\{x_i^j, y_i^j, z_i^j, s_i^j\}_{i=1}^{N_i}\}_{j=1}^{l}$

1: Convert the G_{input} to G_{output}^{l} by a direct replication: $G_{output}^{l} = \{x_i^l, y_i^l, z_i^l, s_i^l\}_{i=1}^{N_l},\ N_l = N$.

2: **Calculate the shifted spatial coordinates:** $(\bar{x}_i^j, \bar{y}_i^j, \bar{z}_i^j) = (x_i^j \ll 2, y_i^j \ll 2, z_i^j \ll 2)$. We treat voxels that share the same $\bar{x}, \bar{y}, \bar{z}$ value as a cell.

3: **Conduct semantics voting** among the voxels inside a cell to determine the most prevalent semantics s^* and its corresponding vote count cnt.

4: **Determine the distance range** in which each cell resides based on its spatial coordinates and obtain its corresponding rule parameter δ^*, $\delta^* \in \{\delta_1, \delta_2, \delta_3\}$.

5: **Merge small voxels within the cell:** If the vote count $cnt > \delta^*$, and s^* is not the ambiguous semantics, the semantics of the cell is set to s^*. Qtherwise, the cell is assigned the ambiguous semantics.

6: **Make the Rule stricter:** $\{\delta_1, \delta_2, \delta_3\} = \{\delta_1+1, \delta_2+1, \delta_3+1\}$.

7: If reached the first layer, go to the procedure 8; If not, go back to the procedure 2.

8: **Perform a reverse elimination process**.

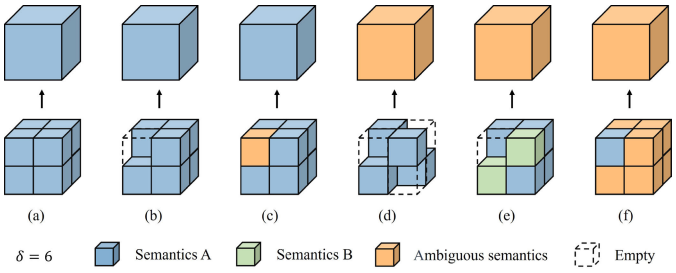


Fig. 3. **Cases of merging small voxels within the cell.** In cases (a), (b), and (c), where the count of the most prevalent semantics A exceeds the rule parameter δ^*, these small voxels can be merged into one larger voxel, and thus, the cell is assigned the semantics A. In cases (c), (d), and (e), where the count of the most prevalent semantics is less than the rule parameter δ^* or it is the ambiguous semantics, the cell is assigned the ambiguous semantics.

status. Finally, when we have completed the construction of all layers, we need to perform a reverse elimination process. This implies that voxels identified as determined voxels in a given layer should not be present in subsequent layers, as they have already been output at that specific layer of the network and are not involved in further computations. The final construction results are shown in Figure 4 and we utilize each layer of the constructed labels to supervise the corresponding layer in the adaptive octree-based module.

E. Adaptive Loss Based on the F_1 Score

We construct four layers of supervised labels based on the aforementioned octree construction rule, ranging from a voxel granularity of 2 meters down to 0.25 meters. And then we compute the proportion of voxels for different semantic calsses at each level as shown in Table I. Clearly, the class imbalance issue remains significant in our hierarchical representation, which affects the overall performance of the network to a certain extent. To address this, we propose an adaptive loss based on the F_1 score, building upon the scene-class affinity loss [20]. Considering p_i the ground truth class of voxel i, and $\hat{p}_{i,c}$ its predicted probability to be of class c, we define:

$$P^c(\hat{p}, p) = log\frac{\sum_i \hat{p}_{i,c}[\![p_i = c]\!]}{\sum_i \hat{p}_{i,c}} \tag{4}$$

$$R^c(\hat{p}, p) = log\frac{\sum_i \hat{p}_{i,c}[\![p_i = c]\!]}{\sum_i [\![p_i = c]\!]} \tag{5}$$

$$S^c(\hat{p}, p) = log\frac{\sum_i (1-\hat{p}_{i,c})(1-[\![p_i = c]\!])}{\sum_i (1-[\![p_i = c]\!])} \tag{6}$$

with $[\![\cdot]\!]$ the Iverson brackets. For each semantic class, we maintain an individual F_1 score:

$$F_1^c = \frac{P^c \cdot R^c}{P^c + R^c} \tag{7}$$

Whenever a new sample is trained, we obtain the F_1 score for each class predicted by the model within that sample, denoted as $\overline{F}_1^c$. We then update the F1 score using a moving average approach:

$$F_1^c = \beta F_1^c + (1-\beta)\overline{F}_1^c \tag{8}$$

where β is the decay coefficient. Then, based on the F_1 scores, we calculate the loss weights for each class:

$$w^0, \ldots, w^C = Softmax(\alpha(1-F_1^0), \ldots, \alpha(1-F_1^C)) \tag{9}$$

where α is a hyperparameter that controls the weight distribution. A larger α will allocate greater weight to classes with lower F_1 scores. So the adaptive loss is as following:

$$L_{ada}(\hat{p}, p) = -\frac{1}{C}\sum_{c=1}^{C} w^c(P^c(\hat{p}, p) + R^c(\hat{p}, p) + S^c(\hat{p}, p)) \tag{10}$$

Given that AdaptiveOcc generates multi-layer outcomes, we impose supervision on the network at each level. We use

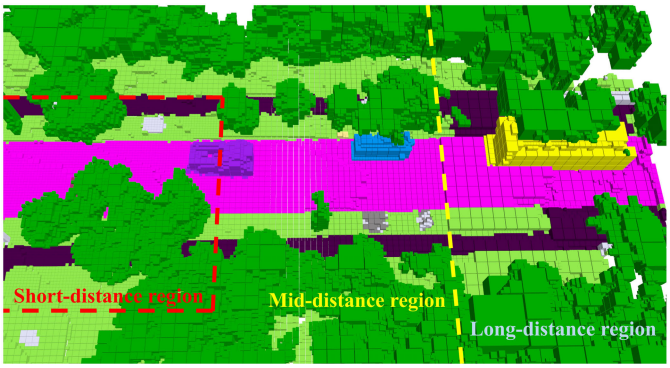


Fig. 4. **Visualization of the constructed supervised labels.** Distance-adaptive octree construction rules results in a higher likelihood of coarser voxel granularity for distant regions and finer voxel granularity for nearby regions.

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TABLE I
PROPORTIONS OF VOXELS FOR DIFFERENT SEMANTIC CLASSES AT EACH LEVEL

level	empty	barrier	bicycle	bus	car	construction vehicle	motorcycle	pedestrian	traffic cone	trailer	truck	driveable surface	other flat	sidewalk	terrain	manmade	vegetation	ambiguity
2m-level (%)	49.391	0	0	0.007	0.027	0.003	0	0	0	0.004	0.015	0.283	0.002	0.037	0.094	0.151	0.613	49.373
1m-level (%)	50.154	0.026	0.002	0.039	0.308	0.030	0.002	0.032	0.003	0.057	0.117	2.576	0.053	0.670	0.887	2.542	2.258	40.246
0.5m-level (%)	56.265	0.119	0.007	0.081	0.841	0.065	0.008	0.126	0.025	0.140	0.274	5.656	0.177	1.933	2.098	5.989	3.733	22.463
0.25m-level (%)	72.121	0.206	0.013	0.131	1.506	0.100	0.014	0.221	0.056	0.201	0.476	7.202	0.311	2.521	2.600	6.968	5.353	0

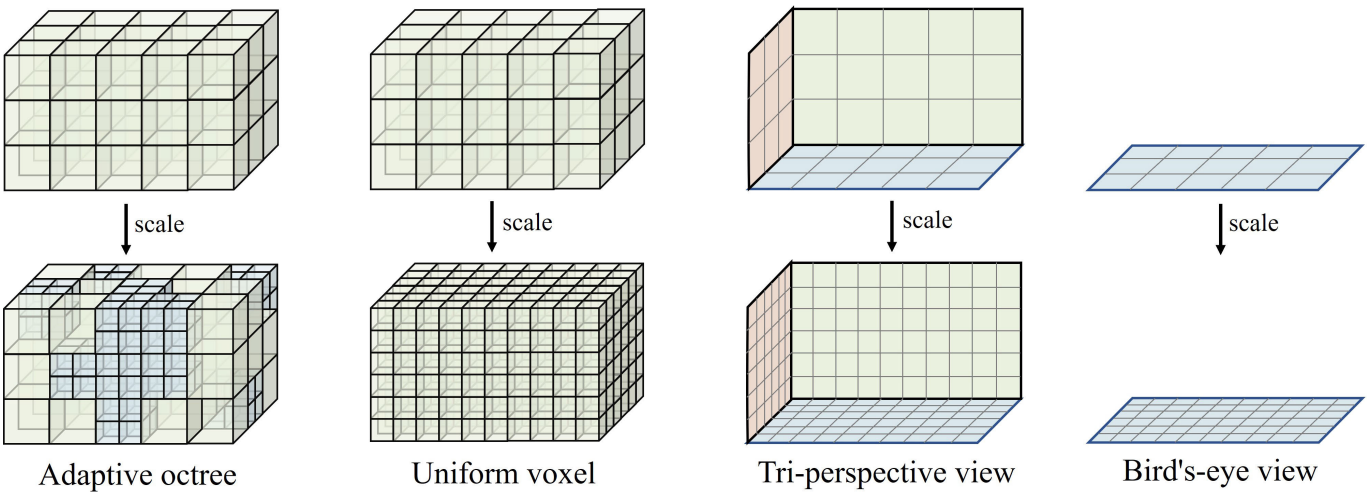


Fig. 5. Comparison of adaptive octree, uniform voxel, tri-perspective view, and bird's-eye view feature representations in resolution scalability.

the same loss function for every layer, which is composed of cross-entropy, adaptive loss, and scene-class affinity loss:

$$L_{total} = L_{ce} + L_{ada} + L_{scal}^{geo} \tag{11}$$

where the cross-entropy loss L_{ce} is the standard loss function for semantic segmentation models, the proposed adaptive loss L_{ada} is employed to mitigate class imbalance, and the scene-class affinity loss L_{scal}^{geo} [20] is designed to be aware of the global occupancy performance.

IV. COMPUTATIONAL COMPLEXITY ANALYSIS IN RESOLUTION SCALABILITY

In this section, we analyze the superiority of our proposed feature representation based on the adaptive octree in terms of resolution scalability. Currently, the feature representation of multi-camera semantic occupancy networks mainly includes three types: uniform voxel representation [20], [21], bird's-eye view (BEV) representation [2], and tri-perspective view (TPV) representation [41]. These feature representations refer to the forms used to encode 3D scene during the process of information propagation within the network.

Uniform voxel representation encodes a 3D scene with dense cubic features $\mathbf{V} \in \mathbb{R}^{H\times W\times D\times C}$,where H, W, D are the spatial resolution of the voxel space and C denotes the feature dimension. This representation maintains the dimensionality consistent with the real world, ensuring sufficient feature expressiveness. When the feature is passed through the network, we denote the storage and computational complexity as $O_V(HWD)$. When doubling the resolution as shown in Figure 5, it changes as:

$$O_V(HWD) \xrightarrow{\text{scale}} O_V(8HWD) \tag{12}$$

BEV representation encodes a 3D scene with a 2D feature map $\mathbf{B} \in \mathbb{R}^{H\times W\times C}$ which compresses the height information. Although BEV greatly reduces the storage and computation burden to $O_B(HW)$, completely omitting the z-axis has an adverse effect on its expressiveness. When doubling the resolution as shown in Figure 5, it changes as:

$$O_B(HW) \xrightarrow{\text{scale}} O_B(4HW) \tag{13}$$

TPV representation uses three distinct 2D feature maps $\mathbf{T} = [T^1 \in \mathbb{R}^{H\times W\times C}, T^2 \in \mathbb{R}^{D\times H\times C}, T^3 \in \mathbb{R}^{W\times D\times C}]$ to encode a 3D scene, derived from three complementary perspectives in space. The storage and computation complexity are proportion to $O_T(HW + DH + WD)$. Intuitively, viewing a complex scene from different perspectives can lead to a better understanding of it, as these perspectives may provide complementary clues about the scene, which has inspired this form of feature representation. When doubling the resolution as shown in Figure 5, it changes as:

$$O_T(HW + DH + WD) \xrightarrow{\text{scale}} O_T(4HW + 4DH + 4WD) \tag{14}$$

Our proposed adaptive octree feature representation can adaptively extend the resolution only on a small portion of

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voxels. For the sake of discussion, we only extend one level of the adaptive octree representation on the uniform voxel representation shown in Figure 5 (actually, multiple levels will be expanded, as seen in Section III, Part B). Consequently, our storage and computation complexity change as follows:

$$O_A(HWD) \xrightarrow{\text{scale}} O_A(HWD + 8N) \quad N \ll HWD \tag{15}$$

where N is the number of voxels that require extending resolution. Statistically, $N < \frac{HWD}{20}$ when scaling from 0.5 meters to 0.25 meters. On one hand, due to the spatial sparsity, a significant proportion of voxels are empty (80% $\sim$ 90%). On the other hand, the distance-adaptive octree construction rules result in certain non-empty voxels not requiring resolution expansion, accounting for about 30% $\sim$ 40%. Compared with equation (12), (13), and (14), our feature representation (15) demonstrates superior capability in terms of resolution scalability. It manages to significantly minimize the increased storage and computational costs brought by resolution scalability.

V. Experiments

A. Dataset

We conduct experiments on nuScenes [24] dataset, which is a large-scale autonomous driving dataset. NuScenes collects 1000 driving scenes of 20 seconds duration each, and the keyframes are annotated at 2Hz. Each sample contains RGB images from 6 cameras with 360 horizontal FOV and point cloud data from 32 beams LiDAR sensor. The total of 1000 scenes are officially divided into training, validation and test splits with 700, 150 and 150 scenes, respectively.

Although nuScenes does not provide official semantic occupancy ground truth, some works [21], [47], [48], [49], [50] have provided methods for generating dense ground truth, as well as the ground truth itself for the nuScenes dataset. However, these works offer ground truths with a singular voxel granularity, and the resolution of these ground truths is relatively coarse ($\geq$ 0.4m). Given that only Wei et al. [21] have open-sourced their method for generating dense ground truth, and can produce at any desired resolution. To validate the effectiveness of our method in resolution scalability, we adopt their approach to generate ground truth with voxel granularities of 0.5m and 0.25m. Same as SurroundOcc [21], since we cannot generate dense occupancy labels in the test set in the absence of 3D semantic and detection labels, we use the training set to train the model and validation set for evaluation. The occupancy prediction range is set as [−50m, 50m] for X, Y axis and [−5m, 3m] for Z axis. The final output occupancy has the shape 200 × 200×16 with 0.5m voxel size and 400 × 400×32 with 0.25m voxel size. The input image resolution is 1600 × 900.

To further demonstrate the effectiveness and generalization of our method, we conduct additional experiments on SemanticKITTI [25] and Waymo dataset [26]. The SemanticKITTI dataset is based on the popular KITTI Odometry Benchmark [51] and focuses on the semantic scene understanding with LiDAR points and front cameras. SemanticKITTI has annotated outdoor LiDAR scans with 21 semantic labels (19 semantics, 1 free, 1 unknown). The ground truth semantic occupancy is voxelized as 256×256×32 grid with 0.2m voxel size. The input image resolution is 1220 × 320. The Waymo dataset does not provide official semantic occupancy ground truth, so we used the Occ3D-Waymo dataset [49] provided by Tian et al. Occ3D-Waymo contains 798 sequences for training, 202 sequences for validation, accumulating 200,000 frames. It has 14 known object classes with an additional general object class. Each sample covers a range of [−40m, −40m, −1m, 40m, 40m, 5.4m], with a voxel size of [0.4m, 0.4m, 0.4m]. The input image size is resized to 960 × 640.

B. Implementation Details

1) Network Structures: We adopt ResNet101-DCN [37], [46] with the initial weight from FCOS3D [53] as the backbone to extract 4 levels' multi-scale image features. These 4 levels' features are fed to FPN for multi-scale fusion, which is then passed on to the adaptive octree-based module. We offer two versions of our model, denoted as AdaptiveOcc and AdaptiveOcc*. The former is trained at 0.5m voxel granularity, using an adaptive octree-based module with 4 layers that starts with an initial granularity of 4m and decreases to 0.5m. The latter is trained at 0.25m voxel granularity, using an adaptive octree-based module with 5 layers that similarly starts at 4m but decreases to 0.25m. We set the number of 2D-3D spatial attention layers to [3, 3, 3] and [4, 4, 5, 5] for AdaptiveOcc and AdaptiveOcc*, respectively. For the distance-adaptive octree construction rule, we set the rule parameters to {6, 4, 3} specifically for short, medium, and long-distance ranges, respectively.

2) Training Setup: We train the model on the NuScenes dataset at 0.5m voxel granularity for 24 epochs and 0.25m for 26 epochs, respectively. We employ the AdamW [54] optimizer, initializing with a learning rate of 2e-4 and applying a weight decay of 0.01. The learning rate was managed by a multi-step scheduler. All models are trained with a batch size of 16 on 2 A100 GPUs with 40G memory.

C. Evaluation Metrics

For 3D semantic occupancy prediction, we use the intersection over union (IoU) to evaluate the occupancy prediction quality regardless of the allocated semantic labels and the mean IoU (mIoU) of all semantic classes to assess the performance of semantic segmentation:

$$\begin{aligned} \text{IoU} &= \frac{TP}{TP + FP + FN} \\ \text{mIoU} &= \frac{1}{C}\sum_{i=1}^{C}\frac{TP_i}{TP_i + FP_i + FN_i} \end{aligned} \tag{16}$$

where TP, FP, FN indicate the number of true positive, false positive, and false negative predictions. C is the number of class. For a fair comparison, we adopt an evaluation approach where the training labels align with the evaluation ground truth to obtain the IoU and mIoU metrics. Specifically, we use 0.5m ground truth to evaluate methods trained at 0.5m granularity

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TABLE II

3D SEMANTIC OCCUPANCY PREDICTION RESULTS ON NUSCENES DATASET AT 0.5M VOXEL GRANULARITY

Methods	Latency (s)	Memory (G)	Range (m)	IoU (%)	mIoU (%)	barrier	bicycle	bus	car	construction vehicle	motorcycle	pedestrian	traffic cone	trailer	truck	driveable surface	other flat	sidewalk	terrain	manmade	vegetation
Bevformer [2]	0.238	13.7	all	30.96	20.31	20.69	12.24	27.57	31.50	10.77	16.76	15.48	14.22	13.39	22.76	38.02	22.13	24.16	21.81	13.10	20.39
			0 ∼ 10	**55.06**	35.94	36.08	31.33	47.84	51.28	20.96	30.94	33.87	28.71	16.93	37.71	63.51	40.26	39.99	36.71	24.18	34.68
			10 ∼ 30	35.26	23.12	24.33	13.38	32.20	34.68	15.22	18.36	17.19	15.10	15.52	25.54	41.56	22.95	27.56	25.52	16.13	24.72
			30 ∼ 50	19.85	11.13	11.05	3.70	15.84	16.65	2.93	9.27	6.12	4.19	9.16	12.84	25.60	12.97	14.84	14.45	7.47	11.05
TPVformer [41]	0.250	18.5	all	31.14	19.97	20.34	12.64	27.43	30.73	10.33	16.62	14.36	12.58	13.05	22.08	38.91	21.16	24.60	22.04	12.42	20.30
			0 ∼ 10	54.99	34.99	34.51	27.49	47.15	50.50	21.65	30.01	32.22	24.53	18.42	37.66	64.01	37.94	40.41	36.91	23.04	33.40
			10 ∼ 30	35.61	22.70	23.84	13.07	31.46	33.99	14.49	18.09	16.03	13.53	14.97	24.76	42.74	22.39	28.09	25.84	15.32	24.68
			30 ∼ 50	19.77	10.66	10.58	3.48	15.92	15.21	2.63	9.20	4.93	2.43	8.93	11.79	26.09	11.99	14.97	14.31	7.09	11.05
OccNet [47]	0.240	14.5	all	32.26	20.44	21.05	11.45	27.88	31.60	10.41	14.71	13.53	11.91	14.43	22.61	39.60	22.13	25.27	23.34	14.84	22.27
			0 ∼ 10	54.96	34.03	35.22	25.85	48.45	50.17	19.25	22.46	30.23	18.05	21.43	37.42	63.37	39.17	39.12	36.95	22.50	34.81
			10 ∼ 30	36.62	23.41	25.06	11.48	32.78	35.02	14.85	16.74	15.25	13.68	16.78	25.41	43.47	23.58	28.65	27.50	17.80	26.45
			30 ∼ 50	21.77	**11.89**	11.26	4.56	16.19	17.24	2.77	8.94	5.23	4.34	9.80	12.66	27.18	13.68	16.65	15.73	9.95	13.99
SurroundOcc [21]	0.256	14.3	all	32.15	**20.65**	21.04	11.27	28.49	31.80	10.69	15.18	14.33	12.25	14.70	22.66	39.69	22.84	25.28	23.75	14.54	21.88
			0 ∼ 10	**55.45**	34.16	35.45	26.77	47.67	50.83	18.54	23.56	31.08	20.58	17.20	37.18	63.75	39.74	39.08	37.30	22.89	34.97
			10 ∼ 30	36.78	23.57	24.88	10.70	33.27	35.28	14.90	17.22	16.05	13.68	17.00	25.46	43.69	24.22	28.87	27.98	17.65	26.19
			30 ∼ 50	20.91	11.80	11.20	4.96	16.53	16.64	3.01	8.91	5.54	4.20	10.28	12.50	26.48	14.01	15.97	15.96	9.33	13.24
AdaptiveOcc (ours)	**0.228**	**13.1**	all	**33.32**	20.11	20.05	10.69	29.06	31.24	10.99	13.25	10.54	8.39	13.17	23.62	40.09	22.79	25.63	23.98	15.52	22.67
			0 ∼ 10	54.81	34.29	35.23	25.21	50.51	53.04	23.39	24.73	24.95	18.46	17.30	39.07	63.71	40.28	40.44	38.14	20.15	34.05
			10 ∼ 30	**39.16**	**23.72**	24.04	11.08	35.88	35.67	16.23	14.88	12.39	8.76	16.36	27.77	45.78	24.70	29.95	28.95	19.20	27.88
			30 ∼ 50	**23.55**	11.11	10.49	1.91	14.52	15.25	2.19	6.05	3.19	1.99	7.36	12.23	28.38	13.96	17.14	16.76	11.13	15.22

TABLE III

3D SEMANTIC OCCUPANCY PREDICTION RESULTS ON NUSCENES DATASET AT 0.25M VOXEL GRANULARITY

Methods	Latency (s)	Memory (G)	Range (m)	IoU (%)	mIoU (%)	barrier	bicycle	bus	car	construction vehicle	motorcycle	pedestrian	traffic cone	trailer	truck	driveable surface	other flat	sidewalk	terrain	manmade	vegetation
Bevformer [2]	0.436	30.3	all	21.48	13.91	12.81	8.67	16.49	20.57	7.87	13.28	11.40	9.14	7.35	14.35	28.60	16.95	17.92	15.08	7.70	14.36
			0 ∼ 10	37.44	25.15	21.67	22.08	29.72	33.22	18.67	24.10	23.18	17.43	12.74	23.75	50.70	30.34	31.64	27.31	12.87	22.93
			10 ∼ 30	23.11	15.54	14.60	10.43	18.97	21.60	11.14	14.34	12.08	9.33	8.43	15.73	30.68	17.21	20.07	17.21	8.96	16.56
			30 ∼ 50	14.77	7.51	6.89	1.40	9.30	11.02	2.10	7.34	4.66	3.20	4.71	7.85	19.57	9.70	10.49	9.51	4.57	7.78
TPVformer [41]	0.521	35.7	all	21.90	14.12	12.87	10.24	16.55	20.62	8.23	13.56	11.08	8.47	7.40	14.66	29.36	17.15	18.43	15.52	7.33	14.40
			0 ∼ 10	37.47	24.92	20.92	20.82	28.69	33.18	19.81	24.19	22.68	15.36	14.03	24.51	50.75	30.09	31.71	27.45	12.25	22.29
			10 ∼ 30	23.61	15.61	14.57	11.35	18.69	21.68	11.49	14.39	11.76	8.68	8.36	16.02	31.60	17.69	20.63	17.61	8.59	16.60
			30 ∼ 50	15.07	7.50	6.84	1.84	9.58	10.54	2.08	7.69	3.95	2.21	4.80	7.67	20.19	9.66	10.84	9.82	4.30	7.92
OccNet [47]	0.534	38.8	all	22.37	**14.62**	13.18	10.78	16.87	21.04	8.30	14.01	11.43	8.95	7.76	15.28	29.57	18.27	18.85	16.25	7.97	15.26
			0 ∼ 10	38.44	25.65	22.39	21.36	28.85	33.89	20.77	24.78	23.74	15.77	14.03	24.55	51.00	31.66	32.36	28.10	12.91	24.23
			10 ∼ 30	**24.22**	**16.33**	15.11	12.16	19.61	22.27	11.88	15.22	12.17	9.40	9.07	16.84	31.85	19.10	21.12	18.54	9.33	17.63
			30 ∼ 50	15.24	7.90	6.91	3.28	9.35	10.92	2.35	7.94	4.45	3.35	4.69	7.98	20.21	10.25	11.17	10.31	4.81	8.38
SurroundOcc [21]	0.551	32.0	all	22.19	14.36	12.98	9.18	17.06	20.80	8.38	12.38	10.57	7.78	8.35	15.17	29.92	18.26	18.92	16.62	8.31	15.15
			0 ∼ 10	37.52	24.18	21.18	20.12	29.34	32.74	16.74	19.82	21.13	12.63	12.45	24.20	50.41	32.15	31.10	27.74	12.00	23.11
			10 ∼ 30	24.06	16.02	14.84	9.81	19.57	22.05	11.61	13.83	11.41	8.44	9.55	16.64	32.25	18.90	21.18	19.13	9.62	17.41
			30 ∼ 50	15.38	8.10	7.00	2.76	9.83	11.05	2.50	7.20	4.08	3.00	5.64	8.28	20.59	10.85	11.56	10.71	5.57	9.04
AdaptiveOcc (ours)	**0.309**	**18.9**	all	**24.36**	14.61	13.11	8.54	18.78	21.80	6.67	11.00	7.50	5.56	7.58	15.47	32.79	19.95	20.42	18.40	9.95	16.28
			0 ∼ 10	**40.51**	**25.83**	24.53	20.07	30.25	38.66	17.64	21.57	19.12	14.93	12.80	27.50	53.87	33.84	33.64	30.36	11.27	23.22
			10 ∼ 30	24.05	15.94	15.00	9.12	22.46	24.02	10.67	12.63	9.16	5.88	8.06	17.62	32.57	20.33	21.34	19.30	8.63	18.18
			30 ∼ 50	**18.00**	**10.76**	9.20	1.92	12.69	14.57	1.73	5.59	2.39	0.94	6.73	10.35	30.25	16.35	18.17	16.80	10.69	13.82

and 0.25m ground truth to evaluate methods trained at 0.25m granularity. Our method is evaluated using the generated multi-granularity supervised labels. Notably, we also partition the entire space into three distinct regions for evaluation based on distance, aligning with our distance-adaptive octree construction rule. On nuScenes dataset, we define the short, mid, and long-distance region as 0∼10m, 10∼30m, 30∼50m (The values indicate absolute distances, meaning they include both positive and negative ranges). We calculate IoU and mIoU for each region as well as the entire area.

D. Main Results

1) Results on nuScenes: We compare our methods with several state-of-the-art methods on nuScenes dataset as shown in Table II and Table III. All these methods are trained individually at a voxel granularity of 0.5m and, after scaling the network, at a granularity of 0.25m. Table II presents the results at 0.5m voxel granularity, while Table III shows the results at 0.25m voxel granularity. It can be observed that our method demonstrates significant advantages in terms of inference latency and training memory consumption. This is attributed to our adaptive multi-layered output architecture, which can significantly save computational and storage resources. Moreover, our method achieves better IoU metrics and competitive mIoU metrics in terms of overall accuracy. And in the region-specific evaluation, our method performs at the state-of-the-art level in more regions. All these comparison methods employ different feature representations. Specifically, SurroundOcc [21] adopts the uniform voxel representation, Bevformer [2] adopts the BEV representation, TPVformer [41] adopts the TPV representation, and OccNet [47] combines BEV and uniform voxel representations. The experimental results in Figure 6 and Figure 7 demonstrate that our method has superior performance in terms of resolution scalability. It can significantly reduce the increased storage and computational costs brought by scaling. Quantitatively, our method exhibits a significant improvement in the growth rates of inference latency and training memory after scaling, with 35.5% and 44.2%, respectively, compared to SurroundOcc's 115.2% and 123.8%, as well as BevFormer's 83.2% and 121.2%. This strongly demonstrates the superiority of our proposed adaptive octree feature representation in terms of resolution scalability,

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TABLE IV
3D SEMANTIC OCCUPANCY PREDICTION RESULTS ON SEMANTICKITTI DATASET AT 0.2M VOXEL GRANULARITY

Methods	Input	Latency (s)	Memory (G)	IoU (%)	mIoU (%)	road	sidewalk	parking	other ground	building	car	truck	bicycle	motorcycle	other vehicle	vegetation	trunk	terrain	person	bicyclist	motorcyclist	fence	pole	traffic sign
LMSCNet [52]	$\hat{x}^{occ}$	0.059	1.7	28.61	6.70	40.68	18.22	4.38	0.00	10.31	18.33	0.00	0.00	0.00	0.00	13.66	0.02	20.54	0.00	0.00	0.00	1.21	0.00	0.00
JS3C-Net [2]	$\hat{x}^{pts}$	0.483	3.2	34.00	8.97	47.30	21.70	19.90	2.80	12.70	20.10	0.80	0.00	0.00	4.10	14.20	3.10	12.40	0.00	0.20	0.20	8.70	1.90	0.30
MonoScene [20]	x^{rgb}	0.332	17.1	**36.86**	11.08	56.52	26.72	14.27	0.46	14.09	23.26	6.98	0.61	0.45	1.48	17.89	2.81	29.64	1.86	1.20	0.00	5.84	4.14	2.25
TPVFormer [41]	x^{rgb}	0.233	17.5	35.61	11.36	56.50	25.87	20.60	0.85	13.88	23.81	8.08	0.36	0.05	4.35	16.92	2.26	30.38	0.51	0.89	0.00	5.94	3.14	1.52
OccFormer [39]	x^{rgb}	0.357	14.5	36.50	**13.46**	58.85	26.88	19.61	0.31	14.40	25.09	25.53	0.81	1.19	8.52	19.63	3.93	32.62	2.78	2.82	0.00	5.61	4.26	2.86
SurroundOcc [21]	x^{rgb}	0.224	12.7	36.42	11.96	57.25	25.68	21.52	0.62	14.51	24.37	8.09	0.52	0.28	6.08	17.99	3.08	31.04	1.74	1.48	0.00	6.75	3.82	2.36
AdaptiveOcc (ours)	x^{rgb}	**0.108**	**7.2**	36.72	12.22	57.05	27.09	21.05	0.83	14.98	25.02	8.77	0.45	0.26	6.99	18.75	3.74	32.27	1.51	1.29	0.00	6.45	3.76	1.88

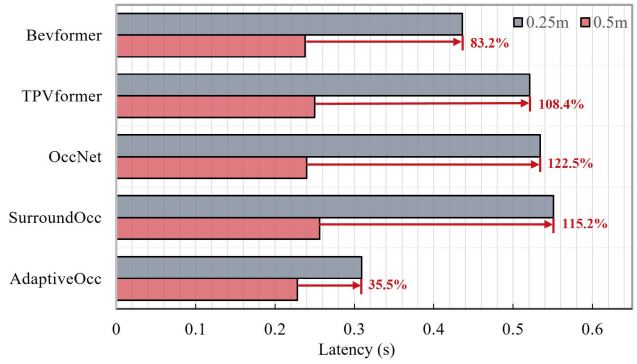


Fig. 6. **Comparison of inference latency gains by scaling.** The relative gains are marked by red.

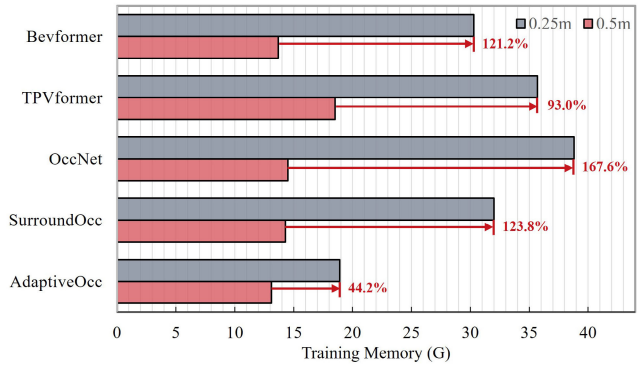


Fig. 7. **Comparison of trainging memory gains by scaling.** The relative gains are marked by red.

aligning with the theoretical analysis in Section IV. We also show some qualitative results in Figure 8, including the ground truth, supervised label, our method, and SurroundOcc. It can be observed that our method possesses adaptive capabilities, with a higher probability of predicting coarse-grained voxels for distant regions and finer-grained voxels for nearby regions. Moreover, compared to the baseline method SurroundOcc, our approach exhibits superior performance in terms of object shape fidelity, particularly evident in the prediction of vehicle voxels.

2) Results on SemanticKITTI: To further demonstrate the superiority and generalization of our method, we also conduct monocular 3D semantic scene completion on SemanticKITTI dataset. We compare our proposed AdaptiveOcc with existing baselines LMSCNet [52] and JS3CNet [2], which respectively use occupancy grids and point clouds as inputs. We also compare our AdaptiveOcc with MonoScene [20], TPVFormer [41], OccFormer [39], and SurroundOcc [21], which are best RGB-only methods. Table IV shows the quantitative results. Although our method is not designed for monocular perception and cross-view attention will be weakened for the monocular setting, our method still achieves competitive performance on this benchmark. Compared to models specifically designed for monocular images such as Monoscene [20] and OccFormer [39], although our model is slightly lower in IoU or mIoU accuracy metrics, it demonstrates significant advantages in terms of inference speed and training memory usage. We are much faster than other monocular methods in inference speed, with the lowest training memory usage. Specifically, our inference speed is more than three times faster than OccFormer [39] with the highest mIoU and Monoscene [20] with the highest IoU. We also show some qualitative results in Figure 9, including OccFormer [39], SurroundOcc [21], our method and the ground truth. It is evident that our method demonstrates a stronger prediction ability for the spatial layout of voxels of various semantic categories, especially for the buildings visualized as yellow voxels.

3) Results on Waymo: We compare the performance of our AdaptiveOcc network with state-of-the-art models on Occ3D-Waymo dataset [49]. Results are shown in Table V. Since most methods implemented on the Occ3D-Waymo dataset are not open-sourced, we only compare the mIoU metric provided by Occ3D-Waymo. Our method achieves competitive performance on this benchmark compared to the state-of-the-art method CTF-Occ [49]. Especially for some categories such as building and vegetation, our method surpasses the state-of-the-art method by 0.64 and 0.66 IoU, respectively. And our method has an inference latency of 0.134 seconds using only 10.3 GB of memory for training.

4) Advantage Analysis: The advantage of our model in reducing inference latency and training memory usage relies on two key factors. First, our model fully leverages spatial sparsity, as indicated by statistics from the nuScenes dataset, where approximately 90% of voxels in the surrounding space are empty. Representing and propagating these empty voxels with high-dimensional features is inefficient and meaningless. Our model can quickly identify and output these empty voxels at each layer, avoiding further computational and storage overhead. Second, our model exhibits distance adaptivity, where the predicted voxel granularity varies with distance. Close-range perception requires fine voxel granularity, whereas the granularity requirements are relaxed for distant areas. Therefore, our model tends to predict more large voxels in

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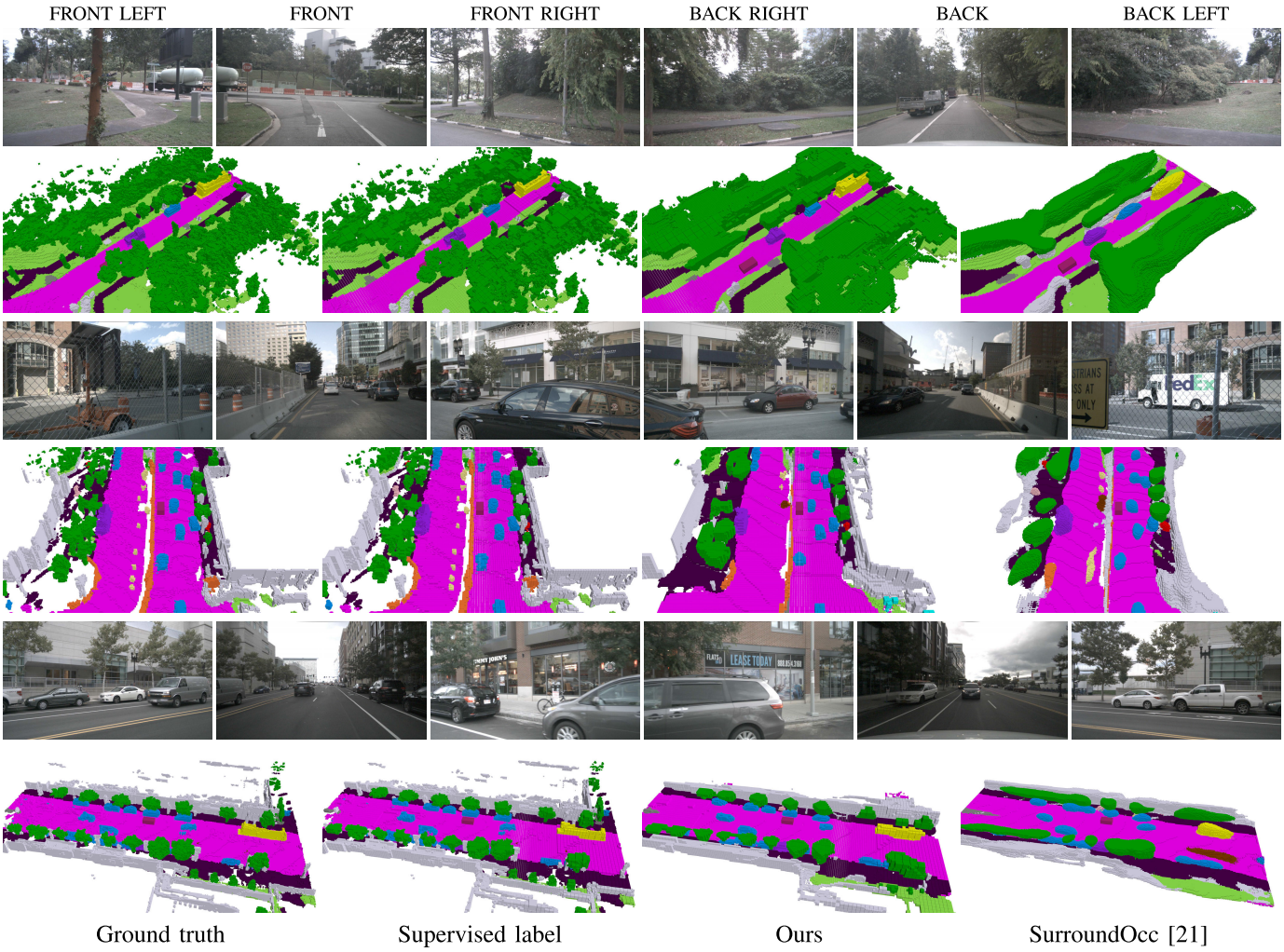


Fig. 8. **Qualitative comparison on nuScenes validation set at 0.25m voxel granularity.** Our method AdaptiveOcc can adaptively predict different parts of space with varying voxel granularities. Compared to the baseline method SurroundOcc, it also exhibits superior performance in terms of object shape fidelity, particularly evident in the prediction of vehicle voxels. The dark red rectangular in these figures represents the ego vehicle. **Better viewed when zoomed in.**

distant regions and outputs them at shallow layers, saving computational and storage resources for subsequent stages.

5) Disadvantage Analysis: Although our model has significant advantages in reducing inference latency and training memory usage, there are some areas where it may not perform as well as the aforementioned methods like Bevformer [2] and OccNet [47]. One of the main drawbacks of our algorithm is its relatively lower accuracy in semantic occupancy prediction for small objects compared to Bevformer [2] and OccNet [47], such as pedestrian and traffic cone categories in Table II and Table III. This is due to the fact that small targets are often treated as indeterminate voxels in the shallow layers of our model because the coarse voxel granularity in these layers cannot finely represent small objects. Consequently, the model struggles to capture the semantic distribution of these small targets in the shallow layers, leading to potential prediction errors. In our latest work, we have addressed this issue by employing auxiliary channel supervision. Please stay tuned for our future work. Our model also faces some challenges in actual deployment due to the use of the octree structure. Compared with other methods, our model introduces control flow and octree operators, which brings certain difficulties to deployment.

E. Ablation Studies

The ablation is conducted on nuScene validation set at 0.25m voxel granularity and from four perspectives: the adaptive loss, the muilti-level architecture, the double skip connection, and the distance-adaptive rule parameters. We use 0.25m ground truth for evaluation and obtain IoU and mIoU for each experimental group. To save training time, the model used in the ablation experiments is lighter than the AdaptiveOcc model in Table III. We set the number of 2D-3D spatial attention layers to [3, 3, 3, 3].

1) Adaptive Loss: We conduct ablation experiments to demonstrate the effectiveness of the proposed adaptive loss. Since our adaptive loss builds upon the scene-class affinity loss, the key difference is that our adaptive loss assigns more weight to minority and challenging semantic classes. Therefore, in Table VI, we compare the network's performance using the scene-class affinity loss and the adaptive loss under different hyperparameters. The experimental results indicate that a reasonable hyperparameter configuration can enhance

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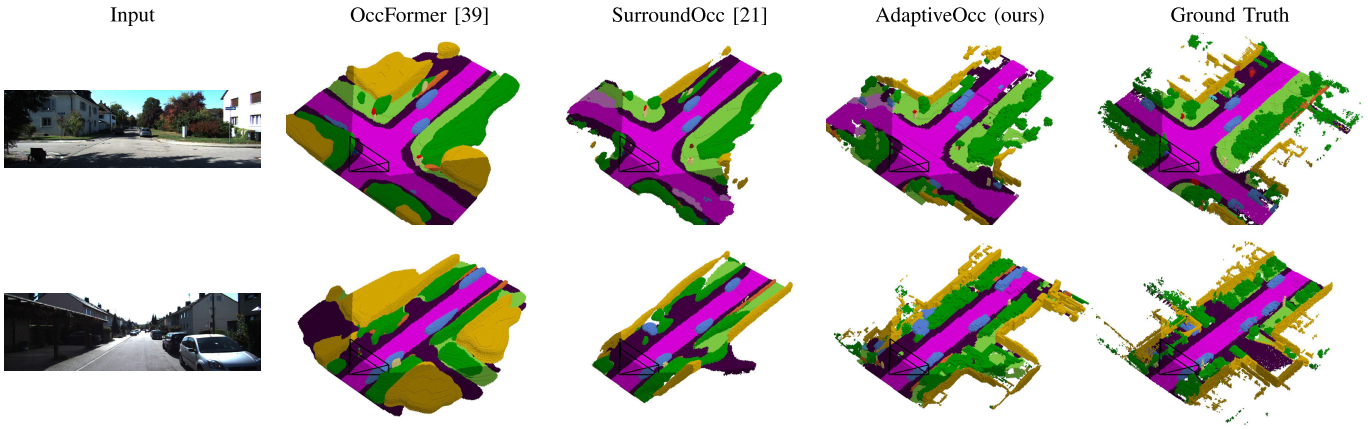


Fig. 9. **Qualitative results on SemanticKITTI validation set.** The input monocular image is shown on the left and the 3D semantic occupancy results from OccFormer [39], SurroundOcc [21], our AdaptiveOcc, and the ground truth are then visualized sequentially. The darker colors within the occupancy images represent the unseen parts out of the camera FOV.

TABLE V
3D SEMANTIC OCCUPANCY PREDICTION RESULTS ON OCC3D-WAYMO DATASET AT 0.4M VOXEL GRANULARITY

Methods	**mIoU (%)**	general object	vehicle	bicyclist	pedestrian	sign	traffic light	pole	construction cone	bicycle	motorcycle	building	vegetation	tree trunk	road	sidewalk
BEVDet [55]	9.88	0.13	13.06	2.17	10.15	7.80	5.85	4.62	0.94	1.49	0.0	7.27	10.06	2.35	48.15	34.12
TPVFormer [41]	16.76	3.89	17.86	12.03	5.67	13.64	8.49	8.90	9.95	14.79	0.32	13.82	11.44	5.8	73.3	51.49
BEVFormer [2]	16.76	3.48	17.18	13.87	5.9	13.84	2.7	9.82	12.2	13.99	0.0	13.38	11.66	6.73	74.97	51.61
CTF-Occ [49]	**18.73**	6.26	28.09	14.66	8.22	15.44	10.53	11.78	13.62	16.45	0.65	18.63	17.3	8.29	67.99	42.98
AdaptiveOcc (ours)	<u>18.54</u>	5.43	27.05	13.16	6.38	14.07	9.37	10.31	13.60	14.38	0.53	19.27	17.96	7.07	71.27	48.19

TABLE VI
ABLATION STUDY FOR ADAPTIVE LOSS

loss	**IoU (%)**	**mIoU (%)**	barrier (0.445%)	bicycle (0.028%)	bus (0.466%)	car (2.970%)	construction vehicle (0.265%)	motorcycle (0.044%)	pedestrian (0.386%)	traffic cone (0.079%)	trailer (0.337%)	truck (1.200%)	driveable surface (24.586%)	other flat (1.072%)	sidewalk (8.027%)	terrain (10.850%)	manmade (22.089%)	vegetation (27.153%)
Scene-class affinity loss	18.13	11.94	10.38	8.47	16.32	18.11	**8.09**	9.47	6.85	5.62	6.05	13.16	**22.90**	17.20	**15.39**	13.06	**6.66**	**13.42**
Adaptivte loss ($\beta = 0.95, \alpha = 1.0$)	**18.35**	**12.22**	**11.31**	**9.45**	**16.40**	**18.12**	8.07	**10.60**	**7.44**	**5.78**	**6.52**	**13.31**	22.63	**17.52**	15.19	13.54	6.58	13.11
Adaptivte loss ($\beta = 0.95, \alpha = 0.5$)	18.21	12.13	11.05	9.45	16.04	18.29	8.05	10.56	7.92	5.52	6.00	12.78	22.44	18.17	14.80	13.25	6.63	13.16
Adaptivte loss ($\beta = 0.95, \alpha = 2.0$)	18.02	12.01	11.49	8.79	16.36	17.97	8.56	10.72	7.66	4.88	6.46	13.35	22.23	17.40	14.79	13.17	6.16	12.17
Adaptivte loss ($\beta = 0.99, \alpha = 1.0$)	18.27	12.17	11.23	9.52	16.02	17.68	8.28	10.53	7.37	6.12	6.59	13.21	22.27	17.60	14.94	13.51	6.50	13.37
Adaptivte loss ($\beta = 0.99, \alpha = 0.5$)	18.22	12.10	11.33	9.00	16.36	18.32	8.17	10.95	7.70	5.55	5.60	12.43	22.23	18.06	15.07	13.54	6.27	12.97
Adaptivte loss ($\beta = 0.99, \alpha = 2.0$)	17.97	12.04	11.45	9.18	16.37	18.27	8.54	11.32	7.35	5.02	5.97	13.32	21.86	17.53	14.87	13.00	6.46	12.16

the network's semantic perception and alleviate the issue of class imbalance. The hyperparameter α has a certain degree of influence on the network's performance, with a larger α resulting in too much weight allocated to minority and challenging semantic classes, causing insufficient learning for normal semantic classes. Conversely, a smaller α fails to effectively mitigate the issue of class imbalance, leading to a decrease in the learning effectiveness for minority and challenging semantic classes.

2) Muilti-Level Architecture: We conduct architecture ablation as shown in Table VII with the aim of finding the optimal layer structure. We set the number of layers in the adaptive octree-based module to be 3, 4, 5, and 6, with initial voxel granularities of 1m, 2m, 4m, and 8m respectively. The experimental results indicate that the five-layer structure consumes the least inference time while maintaining high accuracy. This is because structures with fewer layers do not sufficiently exploit the spatial sparsity and hierarchical representation, leading to a longer inference time. On the other hand, structures with more layers, due to their excessively large initial voxel granularity, fail to leverage the parallelism capabilities of the GPU, resulting in a slight increase in inference time.

3) Double Skip Connection: We carry out an ablation study on the double skip connection in the adaptive octree-based module, as detailed in Table VIII. In this experiments, we

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TABLE VII
ABLATION STUDY FOR ARCHITECTURE

Architecture	IoU (%)	mIoU (%)	Latency(s)	Memory (G)
Three-level	18.07	12.03	0.386	21.43
Four-level	**18.41**	**12.27**	0.321	19.91
Five-level	18.35	12.22	**0.275**	16.31
Six-level	17.83	11.96	0.282	**15.00**

TABLE VIII
ABLATION STUDY FOR DOUBLE SKIP CONNECTION

Method	IoU (%)	mIoU (%)
w/o skip connection	17.55	11.61
single skip connection	18.05	11.98
double skip connection	**18.35**	**12.22**

TABLE IX
ABLATION STUDY FOR DISTANCE-ADAPTIVE RULE PARAMETERS

rules	IoU (%)				mIoU (%)				Latency(s)
	all	$0 \sim 10$	$10 \sim 30$	$30 \sim 50$	all	$0 \sim 10$	$10 \sim 30$	$30 \sim 50$	
$\{8, 6, 5\}$	17.75	35.55	19.72	12.55	11.44	21.62	12.80	6.08	0.305
$\{7, 5, 4\}$	17.93	36.31	20.23	12.67	11.63	22.11	13.26	6.22	0.292
$\{6, 5, 4\}$	18.08	36.78	20.47	12.73	11.77	22.47	13.44	6.38	0.287
$\{6, 4, 3\}$	**18.35**	37.24	**21.11**	**12.94**	**12.22**	22.89	**14.10**	**6.86**	0.275
$\{5, 4, 3\}$	18.21	37.87	20.82	12.87	12.02	23.27	13.77	6.73	0.273
$\{5, 3, 3\}$	17.80	**38.32**	19.43	12.82	11.95	**24.21**	13.18	6.72	**0.272**

compared the network performance between architectures with a single skip connection and that without any skip connections. Specifically, the single skip connection refers to the absence of the dashed structure in the 2D to 3D module depicted in Figure 2, where the module does not receive queries from the previous layer. The experimental results indicate that double skip connections can enhance performance. The explanation is that the double skip connections endow fine-grained features of the current layer with the coarse-grained features from the previous layer, greatly enhancing the expressiveness of the features. Additionally, this residual structure [46] also boosts the learning capability of the entire network.

4) Distance-Adaptive Rule Parameters: Different distance-adaptive rule parameters will alter the supervised labels used to train the network, thus affecting the network's performance. Therefore, we conduct ablation experiments for various rule parameters. We configured six sets of rule parameters, as outlined in Table IX, with rules progressively transitioning from strict to relaxed constraints. Table IX demonstrates that although strict rules($\{8, 6, 5\}$, $\{7, 5, 4\}$, $\{6, 5, 4\}$) lead to a closer alignment between supervised labels and the ground truth, the network's performance, in contrast, experienced a decline. The potential reason is that such strict rules reduce the probability of merging small voxels into larger ones, hindering the full exploitation of the hierarchical structure in our network. Consequently, a substantial portion of voxel predictions occurs in the deeper layers of the network, resulting in increased learning pressure on the network and, simultaneously, increased inference times. On the contrary, relaxed rules ($\{5, 4, 3\}$, $\{5, 3, 3\}$) reduce a little inference time but at the expense of lower accuracy for medium-distance range. Therefore, selecting an appropriate set of rules that balances accuracy and speed is crucial.

VI. CONCLUSION

In this paper, we propose AdaptiveOcc, a novel octree-based multi-level architecture designed for multi-camera 3D semantic occupancy prediction. The adaptability of AdaptiveOcc is highlighted by its capability to intelligently output large homogeneous voxel blocks at shallower layers, preserving computational and storage resources. Simultaneously, the model ensures the propagation of more intricate voxel structures at deeper layers, enhancing fine-grained perception of critical regions. Moreover, to endow our model with adaptability, we introduce a distance-adaptive octree construction rule to generate supervised labels. Such rules result in a distribution of voxel granularity characterized by "fine-grained in the nearby region and coarse-grained in the distant region", which enables the allocation of more computational resources to occupancy prediction in the nearby region. This approach ensures a more efficient and judicious allocation strategy, optimizing the utilization of resources.

Through theoretical analysis and extensive experiments on the nuScenes dataset, we have demonstrated the superiority of our method in terms of resolution scalability. In future work, we plan to apply this method to real driving scenarios, achieving the migration and deployment of our algorithm across various quantities and types of camera devices.

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